MIL-E-2053D

30 December 1970
SUPERSEDING
MIL-E-2053C
8 June 1967

MILITARY SPECIFICATION

EXTRACTOR, CITRUS FRUIT JUICE, HAND

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of hand-operated, citrus fruit juice extractor with base for counter use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-636 - Box, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specification and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 7330

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Traffic Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

NATIONAL SANITATION FOUNDATION (NSF)

Basic Criteria C-2 - Evaluation of Special Equipment and/or Devices.

Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, P. O. Box 1468, Ann Arbor, Michigan 48106.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product.- The extractor delivered under this specification shall be the manufacturer's standard commercial product except for any changes necessary to comply with specification requirements.

3.2 Standards compliance.- The extractor shall conform to the applicable requirements of NSF Basic Criteria C-2.

* 3.2.1 NSF certification.- The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the juice extractor he proposes to furnish under this specification meets the applicable requirements of NSF Basic Criteria C-2 as follows:

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(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF Seal on the finished juice extractor shall be submitted prior to approval of the first shipment, or

(2) A certification for the juice extractor issued by NSF under their special one-time contract evaluation/certification service shall be submitted prior to approval of the first shipment, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the juice extractors are being procured, indicating that the juice extractors have been tested and conform to NSF Basic Criteria C-2 shall be submitted prior to commencing production.

3.3 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial juice extractor provided the completed item complies with all provisions of this specification.

3.4 Design.- The juice extractor shall be designed for counter or table use. The extractor shall consist of the following parts and assemblies: Handle, rack plunger and pinion assembly, removable strainer, and base. The extractor shall operate smoothly and efficiently, with all parts and assemblies adjusted and aligned, and all moving parts snug without binding. The extractor shall extract juice from a half section of citrus fruit in the following manner: A downward force shall be exerted manually through a handle to a pinion and rack. The rack shall be an integral part of the plunger. The plunger shall force the fruit over a strainer. The juice shall flow out through the bottom of the strainer and be collected in a receptacle. The extractor shall operate as intended and the collected juice shall contain no seeds, core, section membrane, or albedo, when tested as specified in 4.4.2. The completely assembled extractor shall be capable of withstanding a 50-pound load applied to the handle without showing any visible damage or deformation to any part when tested as specified in 4.4.1.

3.5 Finish.- The juice extractor shall be finished in accordance with the manufacturer's standard commercial practice. Finished surfaces shall be free from scale and corrosion.

3.6 Marking.- The juice extractor shall be permanently marked with the manufacturer's name, tradename, or trademark of such known character as to be identifiable with the manufacturer.

3.7 Workmanship.- Metallic surfaces shall not contain any rough or sharp edges, burrs, slivers, scratches, rough die or cast marks, gouges, or scale. The molded or cast non-metallic surfaces shall be free from blisters, cracks, dirt, and flash, and all gates shall be trimmed flush. The finished juice extractor shall not be fractured, buckled, punctured, scratched, or malformed. Castings shall not be cracked, porous, warped, or show any signs of blow holes or miscasting. Fasteners shall not be missing, broken, stripped, or fractured.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection.- The inspection lot shall consist of all extractors offered for inspection at one time. The sample unit shall be one completely assembled extractor.

* 4.2.2.1 Visual examination.- Examination of the end item shall be made for defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units. The sample unit shall be one completely assembled juice extractor.

* TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified.		X
	Not clean, smooth or free from discoloration rust, or stains.	X	
Enamel components	Any evidence of chipping, blistering, peeling, or area of no film.	X	
Plated components	Any pitting, peeling, blistering or eruption.	X	
Painted components	Any sags, runs, or area of no film.	X	

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* TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general)	Any component missing.	X	
	Any functioning component inoperative (i.e. rack and pinion assembly).	X	
	Any functioning component that requires abnormal pressure to operate.		X
	Any rough or sharp edges, burrs, or slivers; rough die or grind marks, gouges, rust, or scale.	X	
	Any component fractured, bent, buckled, warped, or damaged (metallic surfaces).	X	
	Any blisters, cracks, dirt, or gates not trimmed (non-metallic surfaces).		X
	Castings cracked, porous, or warped.	X	
	Casting miscast or contains blowholes.		X
Base	Post not at right angles to base.	X	
	Post loose in base.	X	
Rack and pinion assembly (rack, pinion shaft, gear housing, and plunger)	Gear bracket loose on post.	X	
	Plunger loose on rack.	X	
Strainer assembly (funnel bracket, extractor funnel, strainer, and drip cup)	Component not as specified.	X	
	Extractor funnel does not fit in funnel bracket.	X	
	Strainer rotates when in normal position.		X
	Strainer not removable.	X	
Handle	Not as specified.	X	
	Does not fit pinion shaft.	X	
	Loose.		X
	Any fastener missing, broken, stripped or fractured.	X	
Identification marking	Missing, illegible, incomplete.		X

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4.2.2.2 Testing of the end item.- The end item shall be tested as specified in 4.4.1 and 4.4.2, using an inspection level of S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

* 4.2.3 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as indicated in table II. The inspection level shall be S-4 with an AQL of 6.5 defects expressed in terms of defects per hundred units. The lot shall be the number of containers offered for inspection at one time. The sample unit shall be one shipping container fully prepared for delivery.

TABLE II.- Examination of preparation for delivery

Examine	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective.
Weight	Weight of contents not as specified.

* 4.2.3.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

* 4.3 Code and standards compliance.- Proof of compliance with National Sanitation Foundation requirements of 3.2 shall be submitted to the Government representative.

4.4 Test.-

* 4.4.1 Load test.- The base of the extractor shall be secured and a 50-pound load suspended from the handle when the plunger is resting against the strainer. After 1 minute, the load shall be removed and the extractor shall be examined for compliance with 3.4 Inability of the extractor to meet the load requirements specified shall constitute failure of this test.

* 4.4.2 Performance test.- A half-dozen citrus fruits shall be halved and each half shall be inserted in the extractor and the juice extracted. The The operation of the extractor and the collected juice shall be examined for compliance with 3.4. Any nonconformance with the specified requirements shall constitute failure of this test.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Each extractor shall be packaged in a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, class domestic, grade 200 of PPP-B-636. The extractor shall be provided with a folded, die-cut sheet of corrugated fiberboard, as specified for the box, designed to immobilize and prevent subsequent damage to the extractor within the box during transit. Closure shall be in accordance with the appendix of PPP-B-636.

5.1.2 Level C.- Extractors shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Extractors, packaged as specified in 5.1, shall be packed in the most compact manner in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636. Weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B.- Extractors, packaged as specified in 5.1, shall be packed in the most compact manner in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636. Weight of contents of each shipping container shall not exceed 65 pounds.

* 5.2.2.1 When specified (see 6.2) the shipping container shall be a grade V3c, V3s, or V4s, fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

* 5.2.3 Level C.- Extractors, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

* 6.1 Intended use.- The extractor covered by this specification is used to extract juice from citrus fruit.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B shipment (see 5.2.2.1).

* 6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - SA
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Preparing activity:

Army - GL

Project No. 7330-0361

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Extractor, Citrus Fruit Juice, Hand MIL-E-2053D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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